

GROWTH RATES OF SEQUENCES GOVERNED BY THE SQUAREFREE PROPERTIES OF ITS TRANSLATES

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ABSTRACT. We answer several questions of Erdős regarding sets of natural numbers A whose translates intersect with the squarefree numbers in various specified ways. For instance, we show that Erdős’ “property P ” that all translates $n + A$ contain only finitely many squarefree numbers, can only occur for sets of natural density zero, but the decay rate of the density can be arbitrarily slow. For Erdős’ “property Q ” that $n + a$ is squarefree for infinitely many n and all $a < n$, we show that such sets can have natural density as large as $6/\pi^2$, which is best possible. For Erdős’ “property $\overline{P}$ ” (resp. “property $\overline{P}_\infty$ ”) that $n + a$ is squarefree for infinitely many n and all (resp. all but finitely many) $a \in A$, we show that such sets can have (lower) density arbitrarily close to, but not equal to, $6/\pi^2$.

Assume that $A \subset \mathcal{SF} \cap [x]$ contains, for all $p \leq \log x$, all non-zero residue classes mod p^2 with $|A| < \frac{\log^3 x}{\log \log x}$. Then if $n + a$ is squarefree, n must be divisible by p^2 for all $p \leq \log x$, so that $n \geq x^{2-o(1)}$. Then, for all primes p between $\log x$ and $\frac{3}{2} \log x$ and all residue classes $b_p \pmod{p^2}$, add a squarefree integer $a \in (x, x^{3/2}]$ with $a \equiv b_p \pmod{p^2}$. Then we added fewer than $\frac{2 \log^3 x}{\log \log x}$ new integers, so that the new cardinality of A is less than

$$\frac{\log^3 x}{\log \log x} + \frac{2 \log^3 x}{\log \log x} < \frac{\log^3 x^{3/2}}{\log \log x^{3/2}}.$$

1. BASIC ESTIMATES

From the prime number theorem we have

$$\sum_{x \leq p < 2x} \frac{1}{p^2} = \left(\frac{1}{2} + o(1) \right) \frac{1}{x \log x}$$

for all large x , and thus on summing

$$\sum_{p \geq x} \frac{1}{p^2} = \frac{1 + o(1)}{x \log x} \ll \frac{1}{x} \tag{1.1} \text{?}_{\text{psum}}?$$

2020 *Mathematics Subject Classification.* Primary 11B05; Secondary 11B50, 11N25.

Key words and phrases. Squarefree integers, translates, dense sequences.

for any $x \geq 2$. Similar arguments give the variant bounds

$$\sum_{p \geq x} \frac{1}{p(\log \log p)^2} \asymp \frac{1}{\log \log x} \quad (1.2) \text{ ?\{plog\}?$$

$$\sum_{p \geq x} \frac{\log \log p}{p^2} \asymp \frac{\log \log x}{x \log x} \quad (1.3) \text{ ?\{plog-3\}?$$

$$\sum_{2 < p \leq x} \frac{p}{(\log \log p)^2} \asymp \frac{x^2}{\log x (\log \log x)^2} \quad (1.4) \text{ ?\{plog-2\}?$$

$$\sum_{2 < p \leq x} \log \log p \asymp \frac{x \log \log x}{\log x} \quad (1.5) \text{ ?\{plog-4\}?$$

for any $x \geq 3$.

The following basic probabilistic calculation will be used frequently:

Lemma 1. *Let $b \pmod{W}$, $c \pmod{q}$ be congruence classes with W coprime to q , and let I be an interval of length $L \geq W$. Then, if n is drawn uniformly at random from those elements of $b \pmod{W}$ that lie in I , the probability that n lies in $c \pmod{q}$ is smaller than $\frac{1}{q} + \frac{4W}{L}$.*

Proof. We may assume $L \geq 2W$. We then calculate the difference

$$\frac{|\{n \in I : n \equiv b \pmod{W}, n \equiv c \pmod{q}\}|}{|\{n \in I : n \equiv b \pmod{W}\}|} - \frac{1}{q}$$

and aim to show that this is smaller than $\frac{4W}{L}$. By the Chinese remainder theorem, the numerator on the left is equal to $\frac{L}{Wq} + \delta_1$ for some $\delta_1 \in (-1, 1)$, while the denominator on the left is equal to $\frac{L}{W} + \delta_2$ for some $\delta_2 \in (-1, 1)$. We therefore get

$$\begin{aligned} \frac{\frac{L}{Wq} + \delta_1}{\frac{L}{W} + \delta_2} - \frac{1}{q} &= \frac{\frac{L}{Wq} + \delta_1 - \frac{1}{q}(\frac{L}{W} + \delta_2)}{\frac{L}{W} + \delta_2} \\ &< \frac{1 + \frac{1}{q}}{\frac{L}{W} - 1} \\ &\leq \frac{2W}{L - W} \\ &< \frac{4W}{L}. \end{aligned}$$

□

1.1. **Sufficiently rapidly growing sequences.**

Now we adapt the above arguments to prove Theorem ???. Let $\epsilon \in (0, \frac{1}{5})$ be arbitrary and suppose that $A = \{a_1 < a_2 < \dots\}$ is admissible with

$$a_j \geq \exp((4 + \epsilon)j / \log j) \quad (1.6) \text{ ?\{Ajj\}?$$

for infinitely many j .

Let j be a sufficiently large integer for which (1.6) holds, and set $x = a_j$. Then we claim that $j < (\frac{1}{4} - \frac{\epsilon}{2}) \log x \log \log x$. Indeed, assume $j = c \log x \log \log x$ for some $c \geq \frac{1}{4} - \frac{\epsilon}{2}$. If $c > 1$, then $\log j < 2c \log \log x$ which gives $\exp((4 + \epsilon)j / \log j) > x$, contradicting (1.6). If $c \leq 1$ on the other

hand, we would get $\log j < (1 + \frac{\epsilon}{40}) \log \log x$ if x is large enough, while

$$(4 + \epsilon)c \geq (4 + \epsilon)\left(\frac{1}{4} - \frac{\epsilon}{2}\right) = 1 + \frac{\epsilon}{8} - \frac{\epsilon^2}{2} > 1 + \frac{\epsilon}{8} - \frac{\epsilon}{10} = 1 + \frac{\epsilon}{40}$$

so that, once again, $\exp((4 + \epsilon)j/\log j) > x$. Now, let

$$W := \prod_{p \leq \frac{1}{10} \log x} p^2,$$

then by the prime number theorem we have $W = x^{1/5+o(1)}$ as $x \rightarrow \infty$. By the Chinese remainder theorem and admissibility, A avoids at least one residue class $b \pmod{W}$. Fix such a b , and let n be chosen uniformly at random from elements of the residue class $-b \pmod{W}$ in $[x/2, x]$. By construction, $n + a$ is not divisible by p^2 for any $p \leq \frac{1}{10} \log x$ and $a \in A$; if $a \leq x$ as well then $n + a \leq 2x$, and such numbers are also not divisible by p^2 for any $p > \sqrt{2x}$. By Lemma 1 and the union bound the probability that one of the $n + a$ for some $a \in A$ with $a < x$ is not in $\mathcal{SF}$ is at most

$$(c \log x \log \log x) \sum_{\frac{1}{10} \log x < p \leq \sqrt{2x}} \left(\frac{1}{p^2} + \frac{8W}{x} \right)$$

which by (1.1) can be bounded by

$$(c \log x \log \log x) \left(\frac{4 + \frac{\epsilon}{4}}{\log x \log \log x} + \frac{12W}{x^{1/2}} \right) < 1.$$

We conclude that there exists $n \in [x/2, x]$ for which $n + a \in \mathcal{SF}$ for all $a \in A$ with $a < n$. Sending x to infinity, we see that A has property Q as claimed.

1.2. A subexponential sequence. We modify the previous arguments to prove Theorem ???. Let $c_0 > 0$ be small, let C_0 be sufficiently large depending on c_0 , and write $x_j := \exp(C_0 j / \log(j + 1))$ for all j , so $j \leq c_0 \log x_j \log \log x_j$. We will recursively construct $a_j \in [x_j - x_j^{4/5}, x_j]$ such that

- (i) $a_j, a_j + 1, 2a_j \in \mathcal{SF}$; and
- (ii) $a_i + a_j \in \mathcal{SF}$ for all $i < j$.

For C_0 large enough, the sequence $a_1, a_2, \dots$ will then be increasing and obey all the required properties. The property $a_j + 1 \in \mathcal{SF}$ is not directly needed to obtain the conclusions of Theorem ???, but will be useful in the recursive construction.

Suppose that we have already constructed $a_1, \dots, a_{j-1}$ obeying the required properties. We set

$$W_j := \prod_{p \leq \frac{1}{10} \log x_j} p^2 = x_j^{1/5+o(1)}$$

as before (where $o(1)$ is with respect to the limit $x_j \rightarrow \infty$). We select a_j uniformly at random from elements of the residue class $1 \pmod{W_j}$ in $[x_j - x_j^{4/5}, x_j]$, of which there are $\asymp x_j^{4/5}/W$ many. By construction, $a_j, a_j + 1, 2a_j$ and $a_i + a_j$ for $i < j$ are not divisible by p^2 for any $p \leq \frac{1}{10} \log x_j$ (for $a_i + a_j$ this uses the fact that $a_i + 1 \in \mathcal{SF}$ and $a_j \equiv 1 \pmod{W_j}$); they are also not divisible by p^2 for any $p > \sqrt{2x_j}$. For each $\frac{1}{10} \log x_j < p \leq \sqrt{2x_j}$, the probability that any one of these is divisible by p^2 is $O(1/p^2 + W_j/x_j^{4/5})$, thanks to Lemma 1. Thus the total probability that the properties (i), (ii) are not satisfied by this choice of a_j is

$$\ll (j + 2) \sum_{\frac{1}{10} \log x_j < p \leq \sqrt{2x_j}} \frac{1}{p^2} + \frac{W_j}{x_j^{4/5}}$$

which we can bound using (1.1) as before by

$$\ll c_0 \log x_j \log \log x_j \left(\frac{1}{\log x_j \log \log x_j} + \frac{x_j^{1/5+o(1)}}{x_j^{4/5}} \sqrt{x_j} \right) \ll c_0.$$

For c_0 small enough, this is strictly less than one, giving the claim.

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